

Transformer-XL: Attentive Language Models Beyond a Fixed-Length Context

Zihang Dai^{*12}, Zhilin Yang^{*12}, Yiming Yang¹, Jaime Carbonell¹,
Quoc V. Le², Ruslan Salakhutdinov¹

¹Carnegie Mellon University, ²Google Brain

{dzihang, zhiliny, yiming, jgc, rsalakhu}@cs.cmu.edu, qvl@google.com

Abstract

Transformers have a potential of learning longer-term dependency, but are limited by a fixed-length context in the setting of language modeling. We propose a novel neural architecture *Transformer-XL* that enables learning dependency beyond a fixed length without disrupting temporal coherence. It consists of a segment-level recurrence mechanism and a novel positional encoding scheme. Our method not only enables capturing longer-term dependency, but also resolves the context fragmentation problem. As a result, Transformer-XL learns dependency that is 80% longer than RNNs and 450% longer than vanilla Transformers, achieves better performance on both short and long sequences, and is up to 1,800+ times faster than vanilla Transformers during evaluation. Notably, we improve the state-of-the-art results of bpc/perplexity to 0.99 on en-wiki8, 1.08 on text8, 18.3 on WikiText-103, 21.8 on One Billion Word, and 54.5 on Penn Treebank (without finetuning). When trained only on WikiText-103, Transformer-XL manages to generate reasonably coherent, novel text articles with thousands of tokens. Our code, pretrained models, and hyperparameters are available in both Tensorflow and PyTorch¹.

1 Introduction

Language modeling is among the important problems that require modeling long-term dependency, with successful applications such as unsupervised pretraining (Dai and Le, 2015; Peters et al., 2018; Radford et al., 2018; Devlin et al., 2018). However, it has been a challenge to equip neural networks with the capability to model long-term dependency in sequential data. Recurrent neural networks (RNNs), in particular Long Short-

Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997), have been a standard solution to language modeling and obtained strong results on multiple benchmarks. Despite the wide adaption, RNNs are difficult to optimize due to gradient vanishing and explosion (Hochreiter et al., 2001), and the introduction of gating in LSTMs and the gradient clipping technique (Graves, 2013) might not be sufficient to fully address this issue. Empirically, previous work has found that LSTM language models use 200 context words on average (Khandelwal et al., 2018), indicating room for further improvement.

On the other hand, the direct connections between long-distance word pairs baked in attention mechanisms might ease optimization and enable the learning of long-term dependency (Bahdanau et al., 2014; Vaswani et al., 2017). Recently, Al-Rfou et al. (2018) designed a set of auxiliary losses to train deep Transformer networks for character-level language modeling, which outperform LSTMs by a large margin. Despite the success, the LM training in Al-Rfou et al. (2018) is performed on separated fixed-length segments of a few hundred characters, without any information flow across segments. As a consequence of the fixed context length, the model cannot capture any longer-term dependency beyond the predefined context length. In addition, the fixed-length segments are created by selecting a consecutive chunk of symbols without respecting the sentence or any other semantic boundary. Hence, the model lacks necessary contextual information needed to well predict the first few symbols, leading to inefficient optimization and inferior performance. We refer to this problem as *context fragmentation*.

To address the aforementioned limitations of fixed-length contexts, we propose a new architecture called Transformer-XL (meaning extra long). We introduce the notion of recurrence into our

^{*}Equal contribution. Order determined by swapping the one in Yang et al. (2017).

¹<https://github.com/kimiyoung/transformer-xl>